中南大学 2004-2005 学年第二学期《生物化学》考试试卷

Exam paper for the 2nd term of 2004-2005 at Central South University

Subject: Biochemistry
Specialty：Bioengineering
Course character：Required
Exam mode：Without reference documents but permission of any kind of dictionary
Exam time length：120 minutes

Student Name_____, Register No. _____ Score_____

Part I True or False (0.5 point each， 25 pts)

Please write T (true) or F (false) in the blanket

1. Passive diffusion needs energy ( )
2. Glucose transport in erythrocytes occurs by facilitated diffusion ( )
3. All active transport systems are energy-coupling devices ( )
4. Glycolysis provide the organism not only energy but also some important intermediates ( )
5. TCA cycle is the major way to produce energy by aerobic organisms ( )
6. Both photophosphorylation and oxdative phosphorylation produce ATP ( )
7. ATP is not the major energy form in the living organism ( )
8. ATP, acetyl-CoA and glucose-6-phosphate have high-energy bond. ( )
9. NADH normally occurs in biosynthesis pathway, while NADPH occurs primarily in biodegradation pathway ( )
10. Carbohydrate produced by photosynthesis occurs in the dark period. ( )
11. Lipid bilayer takes part in mass transfer, energy production and signal communication ( )
12. Biochemical reaction has not space but substrate specifications ( )
13. Degradation of lipid can provide much more energy and water than carbohydrates ( )
14. NADPH is necessary for maintaining the reductive environments for blood cells ( )
15. Proline often occurs in a terminal and turn of α-helix of protein ( )
16. Alpha helix and beta-sheet of protein is due to hydrogen bond, but subunits of hemoglobulin are arranged due to hydrophobic force. ( )
17. Change in conformation is normally accompanied by breakage of chemical bond(s) ( )
18. Pre-mRNA transcripted from DNA for prokaryotes normally does not need any processing to form the functional mRNA. ( )
19. D or L configuration is referred to the D- or L- glyceroaldehyde. ( )
20. Collagen molecules have Gly-X-Y repeated units, where X=proline, Y=arginine or asparagine. ( )
21. Folding of proteins usually has a “molten globule” formation stage of the assembled domains. ( )
22. The most common disaccharides include lactose, maltose, sucrose, cellobiose and fructose. ( )

23. Chitin and cellulose cannot digest by human being due to their linear α-glycolinkages. ( )
24. α-pyranoglucose and β-pyranoglucose is two epimers. ( )
25. Galactose, mannose and glucose belong to the same family of sugars. ( )
26. Fibrous proteins are normally hydrophobic and have structural roles. ( )
27. Molecular chaperons can help protein folding. ( )
28. Polysaccharides can be classified into monopolysaccharides and heteropolysaccharides, storage polysaccharides are usually monopolysaccharides, and glycosaminoglycans and proteoglycans are often heteropolysaccharides. ( )
29. The major storage lipids are triacylglycerols. ( )
30. Membrane is a mobile system, and it is sensitive temperature that may lead to phase transition of the membrane. ( )
31. The curve of transport rate versus time for facilitated diffusion is sigmoid. ( )
32. The donor of methyl group in living organisms is typically S-adenosyl methionine. ( )
33. Na^+/K^+ pump needs ATP as energy, while Ca^{2+} pump needs energy from Ca^{2+} ion motive potential. ( )
34. Transformable genetic materials may include DNAs, RNAs and proteins. ( )
35. Phosphate group binds to the 3'- or 5'-ribose of DNA. ( )
36. The higher energy bonds of ATP are α and β phosphoester bonds. ( )
37. Melting temperature of DNA may decrease with increase in GC content. ( )
38. Melting temperature of DNA may increase with increase in electrolyte concentration ( )
39. Many vitamins are usually the precursors of coenzymes or prosthetic groups. ( )
40. Enzymes can accelerate the attainment of the equilibrium position, because it can change the position of equilibrium. ( )
41. The role of catalytic active site of an enzyme is to bind regulator. ( )
42. Feedback regulation is kind of reversible and allosteric inhibition. ( )
43. Allosteric inhibition is cooperative, but it is also irreversible. ( )
44. Allosteric regulation is by way of binding of inhibitors to the regulatory sites of the enzyme. ( )
45. In the nonreversible competitive inhibition, the inhibitor binds covalently to the regulatory site of the enzyme. ( )
46. Reversible competitive inhibition can be done by a substance that has a similar structure to that of substrate, while the reversible noncompetitive inhibition can be regulated by a substance that is similar to the structure of the product of the enzymatic reaction. ( )
47. For the K-type of effectors of allosteric enzyme will lead to change in K_m, and V-type of effectors of allosteric enzyme will result in change in V_{max}. ( )
48. According to the Monod-Wyman-Changeux (MWC) Model, substrate binds only to the relax

state of the enzyme, and such a binding is cooperative among the substrate molecules. ( )

49. Reversible covalent modification of an enzyme is usually done by phosphorylation or dephosphorylation of that enzyme's active site. ( )

50. Activation of proenzyme of zymogen is by way of irreversible modification. ( )

Part II Multiple Choice (1 point each， 15 pts)

Choose the one best answer to each of the following questions.

1. ____ Biochemical synthesis of new cell material is called

a) Metabolism
b) Anabolism
c) Catabolism
d) Synthatabolism

2. ____ TCA cycle is a pathway of

a) anabolism
b) catabolism
c) photosynthesis
d) reductive pathway

3. ____ **Which statement is true?**

a) Coenzymes are generally bound tightly to their respective enzymes.
b) Prosthetic groups are generally bound tightly to their respective enzymes.
c) Coenzymes and prosthetic groups are both bound tightly to their respective enzymes.
d) Coenzymes and prosthetic groups are technically not bound to any enzymes.

4. ____ In an electron carrier system, the net energy change is determined by the difference in reduction potentials between the

a) primary electron donor and the terminal electron donor.
b) primary electron acceptor and the terminal electron acceptor.
c) primary electron acceptor and the terminal electron donor.
d) primary electron donor and the terminal electron acceptor.

5. ____ **The Embden-Meyerhof Pathway is another name for**

a) the citric acid cycle
b) glycolysis
c) electron transport
d) NADH production

6. ____ In glycolysis

a) the crucial product is ATP; the fermentation products are waste products.
b) the crucial product is ethanol or lactate; ATP is a waste product.
c) the crucial product is CO_2; ATP is a waste product.
d) the crucial product is not relevant because glycolysis is not a major pathway.

7. ____ The chemical substance that enters the citric acid cycle for further metabolism

is

a) ethyl alcohol
b) acetyl-CoA
c) adenosine triphosphate
d) pyruvic acid

8. ____ Which of the following occurs in the absence of oxygen?

a) Fermentation
b) Anaerobic respiration
c) Anoxygenic photosynthesis
d) All of the above.

9. **____ How many ATPs result from the transfer of electrons from NADH to O_2 through the electron transport chain?**

a) 1
b) 2
c) 3
d) 6

10. **____ The Calvin cycle**

a) is responsible for the fixation of CO_2.
b) utilizes only ATP.
c) requires phosphofructokinase and pyruvate decarboxylase
d) all of the above

11. ____ DNA is always synthesized in the ____ direction.

a) amino terminal to carboxy terminal
b) carboxy terminal to amino terminal
c) 3' to 5'
d) 5' - 5'
e) 5' - 3'

12. ____ If V_{max}=100μmol/mL sec and K_m=2mM, the velocity of the reaction is _____, when [S]=20mM

a) 108 μmol/mL sec
b) 91 μmol/mL sec
c) 67 μmol/mL sec
d) 145 μmol/mL sec

13. ____ The carboxylation catalyzed by a carboxylase usually needs a prothetic group

a) Fe^{2+}
b) biotin
c) ferrocyanin
d) Thiamine

14. ____ In a typical cycle of beta-degradation of C16 saturated fatty acids, it includes two steps of oxidation, in which the electron carriers are respectively :

a) FAD and NAD^+
b) NAD^+ and FAD
c) FAD and FAD
d) NAD^+ and NAD^+

15. **____In an erythrocyte undergoing glycolysis, what would be the effect of a sudden increase in the concentration of ATP**

a) Glycolysis inhibited
b) glycolysis stimulated
c) fructose-1,6-bisphosphate accumulated
d) acetyl-CoA accumulated

Part III Filling Blankets (0.5 point each, 30 pts)

Please fill each blanket with suitable word(s) either in English or Chinese

1. The two major products of pentose phosphate pathway are (_____) and (_____)
2. The fates of pyruvate produced by glycolysis are : (_____); (_____) and (_____)
3. Besides glycolysis, please list the other two most important pathways: (_____); (_____).
4. All the amino acids except (_____) have both free α-amino and free α -carboxyl groups.
5. Sugar and protein are linked together by N- or O-linkage, where N- comes usually from the amino acid (_____), O- comes often from amino acids (_____) and (_____).
6. The secondary structures in proteins include (_____), (_____), (_____), and (_____), and the main features of supersecondary structures in proteins include (_____), (_____) (_____) and (_____).
7. Proteins can be classified into three global classes based on shape and solubility, they are (_____), (_____) and (_____).
8. α-keratins have α-helix, β-keratins have (_____), and collagen have a typical structure of (_____)
9. The most common disaccharides include lactose, maltose, sucrose and cellobiose, among them the homodisaccharides are (_____) and (_____), and the heterodisaccharides are (_____) and (_____).
10. The typical structures of fatty acids and membrane lipids have a (_____) head and a (_____) tail.
11. The four typical biomacromolecules are proteins, (_____), (_____), and (_____).
12. The three major types of membrane proteins are (_____), anchoring proteins, and (_____).
13. The anchoring proteins usually covalently bind to fatty acids by (_____) linkage or (_____) linkage.
14. The two types of passive transport include (_____) and (_____).
15. The active transport need energy that may be typically from (_____), (_____) or (_____).
16. The typical procedure of information transfer includes (_____),

(_____), and (_____).

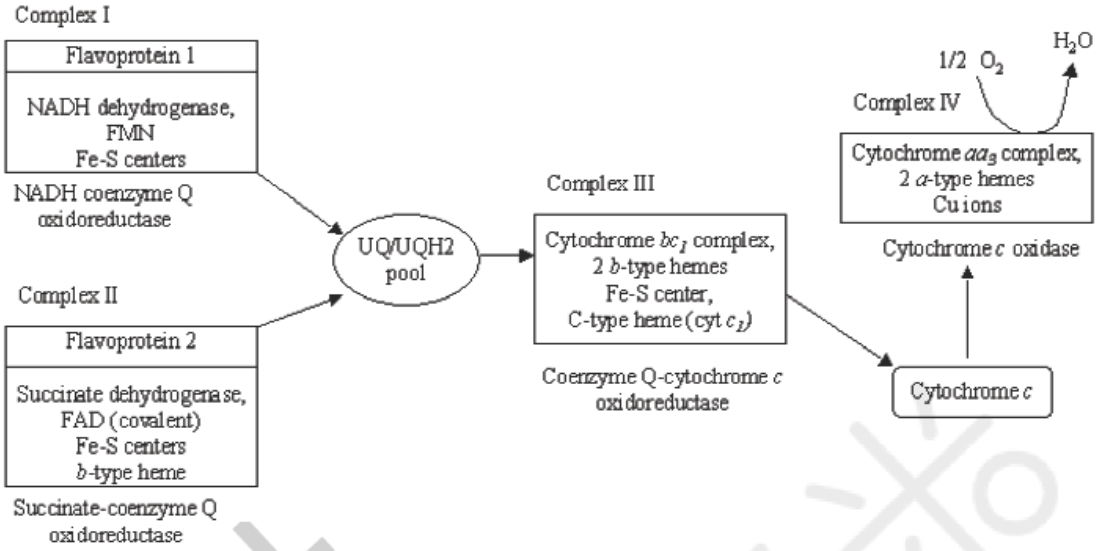
17. DNA may be in form of circurlar or linear, for examples, prokayotic DNA in nucleoid is (_____), while eucaryotic DNA in chromosome (i.e., the genomic DNA of eucaryotes) is (_____).
18. Prokaryotic rRNA contains three types of RNA according to their sedimentation coefficients, they are (_____), (_____) and (_____), respectively.
19. Enzyme cofactors include three types, they are (_____), (_____) and (_____).
20. The binding of a substrate to the active site of an enzyme is by a (_____) force. (choose either covalent or noncovalent)
21. In the reversible competitive inhibition, the inhibitor binds to the (_____) site of the enzyme, and in the reversible noncompetitive inhibition, the inhibitor binds to the (_____) site of the enzyme. Both these two inhibitions are by way of (_____) binding.
22. (Please choose either V_{max} or K_m to fill in the blanket) In the reversible competitive inhibition, the (_____) keeps constant, and in the reversible noncompetitive inhibition, the (_____) is unchanged.
23. In the K-type of inhibition, the apparent K_m will (_____), though the V_{max} keeps unchanged.

Part IV Short essay questions (30 pts)

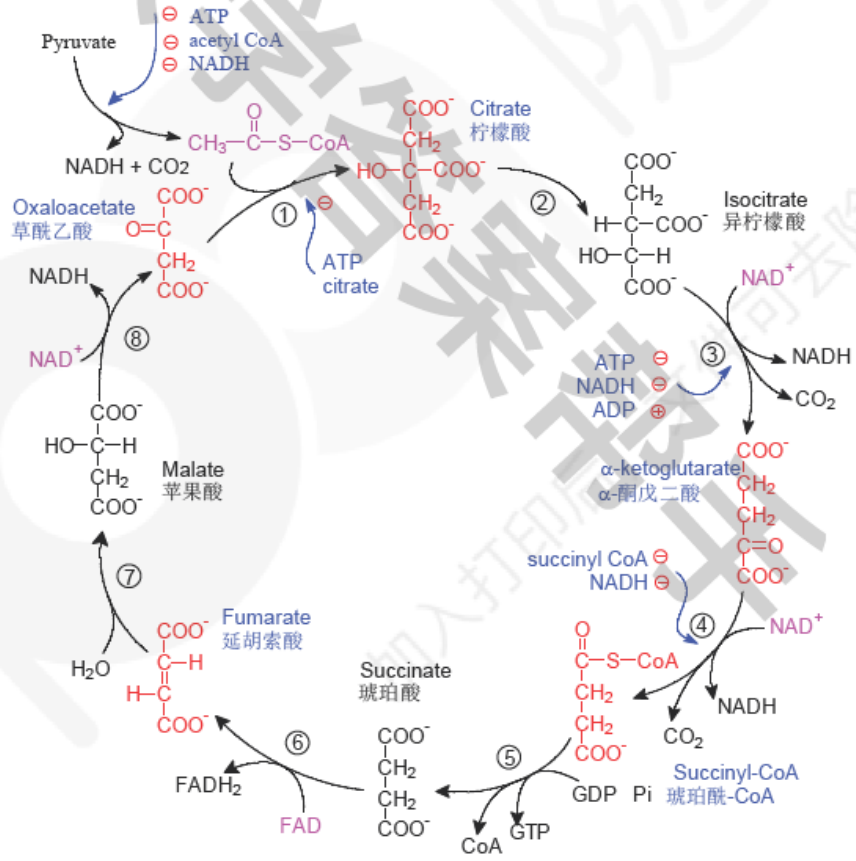
Please answer the following short essay questions either in English or Chinese

1. **Please describe the major points of the chemiosmosis theory. (5 pts)**

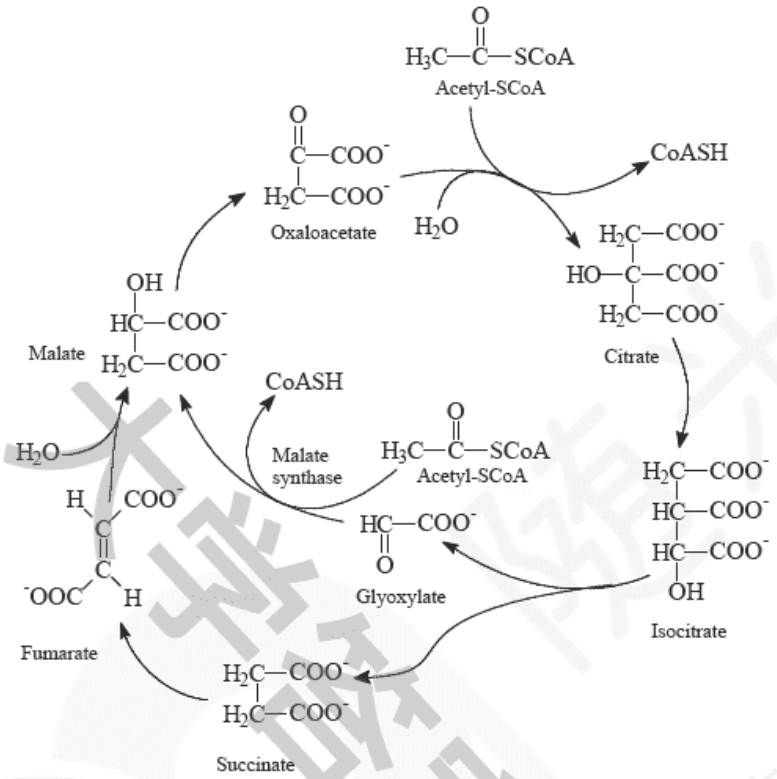
2. **According to the following figure, please point the sources of driving force, the mobile electron carriers, the electron donors and the final electron acceptor. (5 pts)**



3. **According to the following figure of TCA cycle, please point the control steps and how these steps are regulated. (5 pts)**



4. **The following figure is the glyoxylate cycle. Please describe concisely the difference of this cycle from that of TCA cycle, and point which kinds of living organisms will perform such a metabolic pathway. (5 pts)**



5. There are two ways to control the enzyme action, one is to control the enzyme amount that is by way of control of gene expression of that enzyme and it is a slow control way, the other is to control the enzyme activity. Please describe how the enzyme activity is controlled. (5 pts)

6. Why does a deficiency of the enzyme, glucose 6-phosphate dehydrogenase, result in hemolytic anemia?

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参考答案

Subject: Biochemistry　　Specialty：Bioengineering

Part I　True or False　(0.5 point each，25 pts)

1F　2T　3T　4T　5T　6T　7F　8F　9F　10T　11T　12F　13T　14T　15T　16T
17F　18T　19T　20F　21T　22F　23F　24T　25F　26T　27T　28T　29T　30T
31T　32T　33F　34F　35T　36F　37F　38T　39T　40F　41F　42T　43F　44T
45F　46T　47T　48T　49T　50T

Part II　Multiple Choice　(1 point each，15 pts)

1b,　2b,　3b,　4d,　5b,　6a,　7b,　8d,　9c,　10a,　11e,　12b,　13b,　14a,　15a

Part III　Filling Blankets (0.5 point each,　30 pts)

1. NADPH, ribose-5-phosophate
2. To TCA cycle,　To ethanol + CO_2,　To lactate
3. TCA cycle, pentose phosphate pathway
4. proline
5. Asparagine (or Asn), Serine (or Ser), Threonine (Thr)
6. α-helix, β-sheet, β-turn, b-bulge; coiled coil αα, helix bundles αα, hairpins ββ, cross-over βαβ
7. fiberous proteins, globular proteins, membrane proteins
8. β-sheets,　triple α–helix
9. maltose, cellobiose,　　lactose, sucrose
10. hydrophilic, hydrophobic
11. nucleic acids, carbohydrates (or polysaccharides), lipids
12. integrate proteins, peripheral proteins
13. amido. thioether
14. simple diffusion, facilitated diffusion
15. light, ATP, ionic gradient potential
16. DNA replication, DNA transcription, mRNA translation
17. circular, linear
18. 23S, 16S, 5S　(23SrRNA, 16SrRNA, 5SrRNA)
19. metal ions, coenzymes, prothetic groups
20. noncovalent
21. active, regulative (or allosteric), noncovalent
22. V_{max} ,　K_m
23. change, increase

Part IV　Short essay questions (30 pts) (略)

大学答案帮手
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